

Project Title: Amu: A Personalized AI Assistant for Study Mastery

Student Information:

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- Block Section: A

Project Overview:

Provide a concise overview of your project, including its significance and relevance in the field of computer science.

This project centers on developing Amu, an AI-powered study aid designed to enhance students' academic performance through personalized learning tools. Leveraging Retrieval-Augmented Generation (RAG), Amu will enable students to upload study materials (e.g., notes, PDFs, textbooks), which the AI will analyze to deliver tailored assistance, such as summaries, quizzes, and flashcards. By focusing on content-driven support, Amu aims to streamline the study process, offering context-aware recommendations based on uploaded materials. This project contributes to the expanding field of AI-driven educational technologies, addressing the need for integrated, personalized tools that boost comprehension and retention amid growing academic demands.

Problem Statement:

Clearly define the problem your project aims to address. Explain why this problem is relevant and worth investigating.

Students often struggle to process and retain vast amounts of academic material efficiently, relying on fragmented tools like note-taking apps, generic chatbots, or manual study methods that lack personalization. A 2021 study by the National Center for Education Statistics found that 68% of college students report difficulty synthesizing course content, while tools like ChatGPT provide broad but non-specific responses, missing the nuance of individual coursework. Tools like ChatGPT offer generic answers unrelated to specific coursework, while Quizlet requires manual input, lacking dynamic adaptation to user materials. Existing AI solutions rarely integrate user-specific materials for tailored assistance, leaving a gap in academic support. Amu seeks to bridge this gap by developing an AI study aid that uses RAG to analyze student-uploaded materials, offering precise, content-focused tools to improve learning outcomes—a critical need in today's information-heavy educational landscape.

Objectives:

List the specific goals of your project. What do you intend to achieve through this research?

1. Develop an AI assistant capable of generating summaries, quizzes, and flashcards from student-uploaded study materials.
2. Create an AI-powered study space where students can upload academic content for the model to use as a reference for personalized assistance.
3. Implement a RAG-based system to retrieve and generate accurate, context-aware responses from uploaded materials.
4. Provide tailored study suggestions based on the content, structure, and complexity of students' materials.

Literature Review:

Summarize existing research related to your project. Discuss how your project fits into the current body of knowledge and what gaps it aims to fill.

Recent advancements in AI, particularly in natural language processing (NLP) and retrieval-augmented systems, have transformed educational tools. Works like Lewis et al. (2020) demonstrate RAG's ability to combine retrieval and generation for knowledge-intensive tasks, while Brown et al. (2020) highlight the power of large language models in adaptive learning. Existing solutions, such as Quizlet or Grammarly, offer study aids or writing support but lack integration with user-specific academic content. Amu builds on these foundations by using RAG to create a unified, content-driven study aid, filling the gap in personalized, material-specific academic support and enhancing student engagement with their coursework.

Methodology:

Describe the methodology you will use to carry out your project. This section should include your research design, data collection methods, tools, and technologies to be used.

Data Collection: Gather a dataset of academic materials (e.g., sample lecture notes, textbook chapters) from CCS students via free, secure upload interface like GoogleDrive or Mega, ensuring ethical handling and anonymization of personal data.

Model Implementation: Combine a pre-trained language model (e.g., BERT or GPT-3 via Hugging Face) with a RAG framework to process uploaded materials, retrieve relevant sections, and generate study aids. Use Python libraries like Transformers and PyTorch for development.

Testing & Evaluation: Conduct user testing with 10-25 CCS students to assess the assistant's effectiveness. Metrics include accuracy of retrieved answers, quality of generated study aids, and improvement in self-reported comprehension. Iterate based on feedback.

Project Scope:

Define the boundaries of your project. What are the specific features or components you will include, and what will be excluded?

Scope:

Included Features:

- AI-powered study space for uploading and analyzing academic materials.
- Study aids (summarization, quiz generation, flashcard creation) derived from uploaded content.
- Context-aware recommendations tied to material content.

Limitations:

- Offline functionality may be limited due to reliance on cloud-based APIs for processing.
- Precision in retrieving multilingual or highly technical content may vary based on model training data. Initial focus will be on English-language materials common in CCS coursework.

Target Audience:

Identify the primary users or beneficiaries of your project. Who will benefit from your work?

Amu targets CCS students seeking a smart, content-focused study aid to process and enhance their academic materials. By generating personalized study tools and suggestions from uploaded coursework, the app helps students optimize comprehension and retention, benefiting those juggling dense academic workloads and seeking efficient ways to master their subjects.

Expected Outcomes:

State the expected results of your project. Describe what you hope to deliver at the end of your research.

- A user-friendly interface (text-based, with optional voice input) for interacting with the AI assistant.
- A functional RAG-based study aid that achieves at least 70% accuracy in retrieving relevant material and generates study tools rated 4/5 or higher by users.
- Evidence of improved comprehension, with at least 60% of testers reporting better understanding of their materials post-use.

Resources Required:

List any resources, tools, or materials necessary for the successful completion of your project.

- Leverage free tiers of AWS/Google Cloud for initial training and testing, with potential scaling to paid plans if needed.
- APIs for speech recognition and natural language processing (e.g., Google Speech-to-Text, OpenAI GPT).
- Educational datasets for training and testing.
- Development tools (e.g., VSCode, GitHub)

References:

Include any relevant sources, papers, books, or articles that you have consulted during the preparation of your proposal.

- Brown, T., et al. (2020). "Language Models are Few-Shot Learners." Advances in Neural Information Processing Systems.
- Kuhl, P. K. (2010). "Brain Mechanisms in Early Language Acquisition." Neuron.
- Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." Advances in Neural Information Processing Systems.
- Sutskever, I., et al. (2014). "Sequence to Sequence Learning with Neural Networks." Neural Information Processing Systems.
- Relevant educational and AI-focused articles from arXiv and IEEE Xplore.

Conclusion:

Summarize your proposal and highlight the significance of your research.

Amu aims to revolutionize how students engage with academic content by offering an AI-driven study aid that personalizes learning tools based on uploaded materials. Using Retrieval-Augmented Generation, this project addresses the need for tailored educational support, empowering students to master their coursework efficiently. Its significance lies in demonstrating RAG's potential in education, paving the way for smarter, content-focused learning tools.

Signatures:

Student Signature:

Program Coordinator Signature:

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